

Active Inference BookStream #001.08 ~ "Governing Continuous Transform

BookStream | 56:31

hello and welcome

it is January 31st 2023 we're here in

octave bookstream number

1.08 off to you blue

so we will be discussing this book

continuing our discussion of this book

governing continuous transformation by

Jean kesri we are the active inference

Institute

participatory online Institute that is

communicating learning and practicing

applied active inference you can reach

us at all of our social media links this

is a recorded and an archived live

stream so please provide us with

feedback so that we can improve on our

work all backgrounds and perspectives

are welcome here and we will be

following good video etiquette for live

streams

um again if you would like to

participate in this project or any

activities that are ongoing in the

active inference Institute please reach

out to us there are plenty of ways to

get connected and involved

um okay so the goal of today's

bookstream is to continue our discussion

of this book

um and we will just be going through it

as usual um just as always this is not a

review or a final word over this

material just as an introduction to some

of the ideas

um okay so just quick introductions for

those of you who don't know us I am blue
Knight I've been involved in the active
inference Institute for quite a while
I'm an independent researcher in New
Mexico and I'll pass it off

Daniel

hi Daniel I'm a researcher in California
I'll pass to the title

my name is Tyler I'm a critical designer
and GAP researcher in California
right

So today we're moving on to chapter
eight of the book uh the title of this
chapter it's in part two the free energy
governance section and the title of this
chapter is capabilities Duality
management and we'll talk more
um in as we go on as as to what Duality
management kind of means
um okay so there are two sections
outlined by Bijan here in this chapter
there are Dynamic capability View and
antagonistic neural networks perspective
there's a whole bunch of keywords and
we'll just jump right into it um Tyler
do you want to read this part of the
abstract for us

sure all right so abstract Dynamic
capability view GCV and antagonistic
neural networks perspective A and P
address free energy governance
capabilities Dimension the board's
cognitive capabilities in terms of meta
managing
interdependently opposing Logics as
Unity are increasingly relevant to
continuous strategic renewal Duality
management such as the vacillation

between exploitation and adverse
exploration and balancing financials
profusion control is established as a
dynamic board capability effectively FEG
extends dcv to the board level more
importantly by adopting a free energy
principle power perspective by
redefining the firm as an inactive
inference machine dcb's triplet of
sensing seizing and reconfiguring S has
this operationally such that the firm is
engaged as a whole
great so
antagonistic neural network perspective
is established and applied as the
neuroscientific basis for Duality
management managing the existing while
contradicting the present to develop the
future this task positive Network slash
uh what is the other one
gmN Dynamic mode Network I can't
remember uh antagonism what is it
default mode Network there it is
um default mode Network antagonism
between exploitative task execution and
explorative explorative abductive
thinking is a neurological
representation of the contradictory
demands that challenges companies
sustainable outperformance and survival
the board is traditional monitoring and
advisory roles must be explicitly
complemented with The Duality management
mandate
um managing both of these networks more
specifically the board's Dynamic
capability of Duality management lays
the foundation for the board's newly

established governance purpose and
discontinuous and distributive
distributed markets quote Duality
stewardship

um yeah and so we'll get into these task
positive and default mode networks as we
get in later to this presentation
all right

um so this is a dynamic capability view
I looked it up I found this cool image
um so Bijan says sensing strategic
opportunities in Dynamic Market
environments seizing the opportunity and
reconfiguring a firm's resource base
accordingly are a dynamic capabilities
use three Central pillars and he cites
this reference

um so here we see these
fencing seizing and managing or I guess
that's reconfiguring
um

these are the three pillars and then
they give some micro foundations like
what is this sensing seizing
reconfiguration or managing threats
um so I don't know Tyler do you have any
like thing to add here or maybe like
specific examples

um I I mean I see like I I to me this is
a very abstract like diagram and I'm
like what's going on here

yeah I mean I think
this makes it more seem more difficult
and complicated than it really is where
a lot of times with sensing a lot that's
where there's like that abductive
reasoning we're saying like hey there's
something here what could be true here

like how do I explain kind of this like
these disparate signals I'm getting from
the environment how do I make sense of
this and where seizing is like actually
building the thing and managing

Progressive performing is like managing
that thing you built over time and I
think you actually in the next slide it
gives a really good example with um
Amazon Kindle I've actually maybe if you
could just go to the next slide I think
it um explains it well

so for example it says for example
Amazon's launch of the Kindle 2006
involves sensing and the impending
threat of its Core bookselling Business
it's using the opportunity through Rapid
development of the first generation
eBook reader and gradually reconfiguring
its internal activities to ensure that
the Kindle became an integrated part of
its overall Market offering

thanks thank you

um uh bishan goes on to say in a similar
manner to that in which organizational
Learning Resource research distinguishes
between single Loop and double Loop
learning capability research also
distinguishes between operational
capabilities and dynamic capabilities
and so um in the text he gives some
examples of like operational
capabilities like mergers and
Acquisitions and other like things that
are outside the scope of this Dynamic
capability View

um okay so start reading these claims so
he says there has lately been a shift in

Dynamic capability view from behavioral
to cognitive interpretations explicating
the cognitive micro foundation's
underpinning thesis 2007 Three core
pillars of dynamic capabilities sensing
perception seizing sense making and
asset configuration language
communication and social cognition so
this is like reframing again those those
three pillars kind of more towards the
active inference of free energy
governance perspective

um he was on to say with respect to
Dynamic managerial capabilities for
seizing and reconfiguring in particular
we have emphasized ways in which top
management can shape aspects of the
organizational context such as by
encouraging cooperation lowering
resistance to change designing a
business model for a new Venture and
investing in new skills and assets this
suggests that managerial cognitive
capabilities May function as mediators
of the relationship between changes in
organizational context and strategic
change which in turn can affect firm
performance

do either of you guys have any comments
here should we move to the next one okay
all right so I had to look this up
because the term industrial vertical was
like brought up and I was like what is
industrial vertical um and I think like
it's more commonly called an industry
vertical you could have it like in the
industrial field and then it's like
something different but an industry

vertical is also known as a vertical market and it's like a unique business Niche that serves a specific set of consumers or customers and their particular needs and there's like tons of examples of that like I don't know organic food and I don't know like solutions for a biology lab or like there's you know all these like Niche markets and and it's like if you're trying to sell chemicals to a biology lab like you know you're selling them like hplc purified acetic acid and like you're not selling that acetic acid to you know someone to use in their kitchen for salad dressing like it's a totally different like even though you're selling acetic acid those are two totally different markets of people so you don't want to make a product that's like bought by everyone because then it actually is not useful for anyone that was kind of my understanding do you have anything to add Tyler yeah it makes it could give like a little more color of examples here just like an example of an industry vertical could be like movie making business or like on the top end you might have like where the movies are consumed it's a movie theater and then you have how it's produced so you have Studios you have screenwriters uh you're the people who are actually creating the equipment and making the film right so it's basically like this big kind of almost supply chain or these interrelated businesses that are all supporting each other to one

um final product so I one thing I would add here too that I think the Jean would probably agree with is that in a more discontinuous world the idea of an industry vertical is becoming a little bit more fuzzy and so for example like in the past you might say like well I have this I make research I make chemicals for biology research right but it could be the case that you also make chemicals that are used for like hair products and so like your verticals are increasingly becoming more fuzzy and intertwined and it becomes harder to think about like like where do I actually like what Niche do I actually serve like what is the market I'm in right and it could be really hard to think that and that's where sensing really comes in handy is because you could be like oh well I thought I was making biology research chemicals but it turns out because I've I'm sensing and using my adductive reasoning and when they've you know my best fpg perspective I can I'm actually in the hair care market and that's where I really should be um and so it's kind of the idea of a really cool becomes harder to Define uh over time this website actually gave a lot of examples about how uh like industry verticals allow companies that would sometimes be competitors to um collaborate with each other like they talked about how the Samsung glass is used by the Apple iPhone or maybe it's

the other way around but either way like Samsung and like the Samsung phones and the Apple phones are normally like you know competitors in a market um I don't even know that they compete anymore like either you're a Mac person or you're an Android person like I don't think that there's any competition there but maybe

um so but how the you know they're collaborating with the use of the glass and making it accessible to everyone just because everybody needs glass right um okay so they also Bijan also brought up um oh yeah here is the industrial vertical so let me read this from the from the book it says indeed independently of industrial vertical the increasingly distributed and discontinuous nature of Market transforming forces cannot be overestimated however Market velocity velocity is too generic of a concept and so like I was trying to understand um

like how he was using the term Market velocity like here and like so Market velocity like when you Google it it's like it indicates the level of Market activity in individual NASDAQ stocks by measuring the frequency and share volume in aggressively priced orders it can indicate changes in Market Direction momentum or liquidity in each stock enabling users to imply future price movements And Trades so like that's how like Fast the Market's moving I guess

um and then like it's also a company that recycles and like velocity is a measure of two Dimensions right the speed of change and the change in Direction so I don't know

um like the concept of Market velocity here like he can he goes on to compare it to environmental scarcity so the more suitable

um Dynamic capability concept is environmental scarcity uh inspired by Cameron and zamuto 1983 meaningfully relating sensing and sense making as meta capabilities respectively to econation Dynamics and strategic renewal so I don't know um like how environmental scarcity and Market velocity like how those two relate to each other I don't know if you guys can offer any insight here

um but this is something that like I would hold out and ask John about yeah I think with with Market velocity like a lot of times when people talk about discontinuities and like like people who are not coming from an fpg background I'm talking about hey like things are moving really quickly it's more uncertain than ever they talk about Market velocity right like things are just changing all the time and I think what vagina is saying is like just saying that things are changing very quickly isn't very helpful to you because you're like okay cool like life is hard I get it like now what do I do with that information where when you presents environmental scarcity it's

becoming a little bit more specific and like how you can actually be more action oriented with it because you're like okay well like what's my Niche and what are the meta capabilities I have to to create to manage this uncertainty and create my own Niche uh whereas Market velocity is kind of like being overwhelmed by uncertainty and being like I don't know things are hard like this is scary that's at least my take so she goes on to um oh sorry go ahead one related angle to connect back to what we've discussed elsewhere is this concept of generalized coordinates which is the value that something is at and then the first derivative which is the velocity and then the acceleration higher derivatives and so on and so that allows you to track one variable like the temperature or the salinity of the water or the funds in the bank account and look at it not just where it is but how fast it's moving and how fast it's changing how fast it's moving and then you can learn certain things like oh the third derivative is always within this range so that puts certain bounds on timelines that the primary indicator might change by a certain amount or not and we see some of those things like um the gradual decline sudden decline those are differentiated not by their starting point but by the rate of their decline so velocity is something that we can talk about in Bayesian physics actually as a speed

um and then he goes on to outline this environmental scarcity
um from this Schmidt at AI 2016 reference which says combining the type and rate of decline allows us to distinguish four environmental scarcity situations environmental dissolution which is the gradual decline in shape environmental erosion gradual decline in size

environmental collapse sudden decline in shape and environmental contraction sudden decline in size so like I was trying to like picture this as like a like a

um I don't know maybe like a market right like and so like the market changes shape like what would it mean for a market to change shape really um I get like the market changes size like all of a sudden you know there's not a market for CDs anymore because everybody's listening to mp3s so like that's like a

um

well I guess like that's a change in shape and size I'm not sure how to really think so so to use that example like that would be a change in shape where like for example imagine the same and number of people were listening to music but the format changed so it's like the the market is music people who listen to music but the shape is a little bit different of like how they're consuming it whereas it could be a try to change its size and shape as if like you move to mp3s now because it's MP3s

more people can access the music the market grows and that'd be a change of size and shape
or like the market totally disappears because we can just stream whatever we want on YouTube now like yeah exactly that's like that's also a change in size and shape I guess
um but yeah okay cool all right so uh environmental scarcity is more suitable than Market velocity and I guess I can see why that that is because it it's more nuanced as to like what is actually changing is it just like up or down and how fast or like the shape and the size um and like the rate I guess all of that is is taken into this environmental scarcity concept
um okay uh do you want to read these claims Tyler
sure so while slow and gradual Market changes tests in particular our sensing capability more certain changes challenging firms sense making capability uh there's really is no objectively uncertain Market environment uncertainty is an ad for dependent mental experience facing one's very limitations of dynamic AirPlay between sensing and sensor making therefore independent of the degree of Market philosophy Dynamic capabilities are critical he goes on to say the stronger and setting the impact of environmental scarcity the higher the urgency to act and hence the stronger the tendency for strategic renewal to be rather Financial control center I M A and other forms of

alliancing precisely for two reasons one
these the existing resource base is too
trapped in the past to unlearn Advanced
renewal from within and without or to
the opportunity for Market consolidation
is too irresistible and as JP Morgan Had
demonstrated during the 2007-2008
financial crisis and time to action is
critical

cool

um so yeah I guess that that kind of
sheds more light on the environmental
scarcity

um concept here

uh okay so just I wanted to Define this
we kind of talked about the tyranny of
the ore

um last time but we didn't really

elaborate on it and so we we last time

we talked about this good to Great to

Gone book um Circuit City and uh the

tyranny of the ore is actually um Bijan

keeps quoting this wurtzel book about

the tyranny of the Lord but I had to

like look it up it's actually from the

not good to Great Jim Collins book but

like another Jim Collins book that I

haven't read uh built to last and so

here the tyranny of the ore is the

rational view that cannot easily accept

Paradox that cannot live with two

seemingly contradictory forces or ideas

at the same time the tyranny of the ore

pushes people to believe that things

must be either A or B but not both

um and so I think that this concept kind

of goes like is like a light

introduction and we'll dive deeper into

it about like Duality and Duality
management like being able to hold two
simultaneously
um opposing or maybe seemingly opposing
or necessarily contradictory opinions
and reviews
ah okay and here we go right into uh
Duality management
um so Vision says in fact at any point
in time a firm is challenged to balance
strategic organic r d driven Innovation
and growth with financial control
external growth and Innovation by way of
forms of strategic partnering there is
no tyranny of War it is The Duality a
continuous balancing act that is
mutually reinforcing and with
perspective and time and space forms a
Unity free energy governance establishes
Duality management as a dynamic
capability specifically as a dynamic
board capability Duality management such
as vacillation of exploitation versus
exploration and balancing financial and
strategic control is established as a
dynamic board capability the
goal-directed action-centric and
generative circular causality of
top-down bottom-up and lateral sense
making processes prediction error
minimization forms the basis for
leveraging opposing Logics in the form
of duality
um and Duality Management in contrast
with uh strategy management do you want
to read this one Tyler
sure so indeed strategy management has
traditionally embraced one environmental

co-creation versus co-alignment two
exploitation of the first exploration
and three Financial for strategic
control as dualism or Paradox rather
than as Duality it's the background of a
world that is increasingly distributed
just continuous and in this capability
interdependent persistence persistency
and contradiction of interdependent
elements which is the defining core
Paradox it's potentially nonsense and
therefore severely limits the
theoretical as well as practical
relevance of a strategic Paradox lens
foreign

you know the old way and as opposed to
Duality management which is
um the new way

do you guys have anything to add here
this is kind of like Paradox is
attention to balance

Unity is plural and at minimum two
rather than Paradox or something that
must be resolved one way or the other so
it's really compatible with a lot of
qualitative things people say about
wisdom and balance and I think the
question will be how is that enacted in
organizations and how does that connect
to active

yeah I mean one thing that's interesting
about Duality management is that there's
also like a strong cultural tendency to
be decisive or a biased to be decisive
um in that like the perception of kind
of

going not wanting to make a choice can
be seen as weak

um whereas like you haven't really so
like there's even if I think a lot of
people would recognize The Duality and
say like yeah like I don't know it's
kind of both like you want to explore or
you want to exploit like I think a lot
of people would would recognize that but
when you're in a meeting and you're not
like really committing to something well
you're not like really a lot of people
feel like they're not really offering
Like A New Perspective like they're not
putting their stake in the ground saying
like this is where I how I feel about it
and this is where I'm gonna like you
know push behind uh and so I feel like a
lot of people feel this pull to like
make a call and so that's one like even
at an executive level you can feel that
especially two-year like on the
executive level and even if you
recognize this Duality and you're
communicating what you want to do to the
broader organization like Duality is a
really hard thing to communicate and say
like yeah we don't really know what's
going on like it could be one of these
multiple things like let's figure it out
together I have done this look I have
like communicate this before it's A Hard
Road to go down right we're like hey
yeah we're all just figuring it out like
it's gonna be like let's just kind of be
fluid about it a lot of people don't
like to hear that because it sounds like
well this person doesn't know what
they're doing and I don't even like know
where we're going this is really scary

um and so I do want to touch on this
later but like it's one thing
recognizing duality in yourself and on a
small management team how you talk about
duality in a broader organization
there's like a whole different animal
um and so I hope we can talk about that
more a little bit later
yeah thanks for that
uh okay so um we talked last time also
about
um like the quantum Duality and like
what what does patreon mean and like we
knew that he would really kind of
um unwrap it here in this chapter so um
I I guess like we've seen Duality like
we
um are able to like hold simultaneously
it's simultaneously a one and a zero and
so we were wondering if if uh what was
really meant by Duality like in the
quantum sense was more like the
superposition
um angle and so it it seems like yes
um uh based on even this quote from the
book sober Jean says it is important to
appreciate that many dichotomies that
have Divine defined the science and
practice of strategy management such as
firm versus environment executive
leadership versus non-executive Board
individual versus firm exploitation
versus exploration Financial control
versus strategic control formal versus
informal channels and flexibility versus
stability for example are less and less
dichotomous in a distributed and
discontinuous world but coexist in

superposition as Duality or

synchronicity

so I think that this kind of maybe

answers at that question for us from

last week

if you guys have any comments here

okay

so oh Daniel

yeah some of those dualities are raised

as a partition like firm from

environment or organism versus

environment it's a partition that is an

integrated whole and so that is part of

this framing question seeing it as an

integrated unit but also containing

um separate aspects and then also

Quantum carries a lot of baggage whether

somebody's familiar with what it

technically means or not dualities in

the quantum space could refer to wave

particle it could refer to Observer

observed and so I think it needs a lot

more operationalism to have utility

Beyond a sort of like evocative Quantum

management and then again to come back

to Tyler's point which I want to hear

more about

um

if you say well we're actually doing all

the strategies it's like a wave you

thought we were going to go to one

strategy or the other we're going to be

probabilistic across all of them how

does that work

Tyler I'm waiting for your comeback oh

well I mean honestly like

I have actually strategy of how I do

this and I just don't necessarily

communicate broadly that we're doing oh
the strategy is probabilistically like
like I might know that and maybe a
couple other people know that we're like
kind of like keeping our options open
for a bunch of different Avenues but you
communicate something that's much more
clear

um you can't I don't think you get to do
both but you don't get to communicate
that broadly

it's just not consumable by other people
and it doesn't give them like it's not
really actionable by them so you can't
say we're doing all the strategy there
because they're like cool then how do I
fit into this you have to say like well
this is broadly what we're doing and
this is how you're going to fit into it
um and so yeah

also action helps cut this knot so for
any given search space it's not possible
to have one parameter value that is
maximizing exploitation and exploration
that gets into the sort of like I'm
everywhere on the number line at once
type thinking but action allows us to
say well our strategy is we're going to
move quick but as soon as criteria one
two and three are met then we're going
to lock in so that's actually a plan
that you can have Precision about that
embodies in a another variable space
explore exploit but where we have
Precision about
is about action which is what his whole
emphasis is on
actually and that reminds me another

strategy I use too is that you explicitly frame it around exploration and that like you're waiting for something to exploit so you're saying like hey all right we're going to build this thing and we're not sure like who's gonna even use this right but we're able to say we have a hypothesis that these kinds of people might use it and then we're gonna see who uses it and that's like the experiment we're running together as a company is this like Define scope explore stage and then once we run that experiment then we move to exploit and so you can like kind of like clearly do that within an organization of like how you're switching between these different modes

nice

okay uh Bichon says therefore to effectively operationalize Dynamic capabilities triplet of sensing seizing and reconfiguring we need a model that dynamically links Dynamic capabilities triplet with the environment this is free energy governance is underlying and active inference one active States acting sensing two sensory States perception sensing sense making and seizing the opportunity and three internal States firm resources resource configuration and four external States unknown hidden are all Linked In circular causality conditioned by the internal external Markov blanket and governed by free energy minimization and so I just thought it was interesting here like we can go back and look here

at the three pillars like
um the sensing seizing and like
reconfiguring and like for me like the
sensing part
um like goes here like in this action
sensing or sensing is like in sensory
States but like seizing the opportunity
as a sensory State like totally threw me
off

um like is it just me like a sensory
like sensing the opportunity is like one
thing but like leveraging an opportunity
as like a sensory State like that seems
more like an action State I don't know
if it's just me or like what you guys
think about that

um because I was kind of like whoa
seizing the opportunity is a sensory
state

well for example in seizing he included
like creating your business model
and so that is I think part of sense
making because you're like this is how
we relate to the ecosystem so it's like
action like this action centered because
there's also sensory or something like
well this is the kind of like

information that we're going to collect
from the environment as well
and so I think you're kind of doing both
the action and perception
when you're doing season here so like
this is how I'm going to relate to my
environment

um
not sure if that's exactly was what you
were saying I think this is a attempt to
coerce for better and For Worse a

triplet

into the particular partition

where blankets are defined as graph a

Bayesian graph where there are nodes

that make two other sets of nodes

conditionally independent and I think

we've seen in the philosophical and

application settings a lot of challenges

with people simply coercing a

naturalistic understanding of how things

are in the world with this Bayesian

graph formulation like here we have

active States action sensing

action states are the outgoing

statistical dependencies of an entity so

yeah they are action and if you think

about active sensing then they're sort

of elusively related to sensing but it's

not a sense State and so how much are we

going to have both feet in the technical

formalism of the particular partition

and to what extent are we kind of

leaving one foot on the formalism and

then pivoting around and trying to draw

in some other Frameworks and draw

concordances

so I I agree that it's like a coerced

um like like put moving the triplet into

this like particular partition but I I

just don't know if seizing the

opportunity is something that I would

include as a sensory State

um I I think that that's something that

I would put like even in constructing a

business plan like that's still an

action that you'd like Take On The World

so so that's something that's outgoing

not something that's necessarily

incoming and that's just how I would
frame it
um or like the way that made more sense
to me so I just found this a particular
like a particularly curious

um area to to put it
um but yeah we can move on and talk
about we can ask me John why he put it
there uh hopefully and then we can talk
about this antagonistic neural networks
perspective

um so Bhajan says antagonistic neural
network perspectives is established and
applied as the neuroscientific basis for
Duality management managing the existing
while contradicting the present to
develop the future it is the form of
organizational learning that Strategic
Management literature has labeled
ambidexterity co-managing exploitation
and exploration

in neuroscientific terms the
antagonistic relationship is between two
large-scale cortical networks the task
positive Network tpn and the default
mode Network dmn which simply put our
neuroanatomical representations of top
Management's day-to-day task specific
focus on exploitation the task positive
Network on the one side and the rather
non-routine Divergent and creative
thinking that initiates exploration and
strategic renewal the default mode
Network

um and so I thought this was super
interesting and took like a deeper dive
into like looking up the reference for
that uh quote

um and so here in this paper
antagonistic neural networks underlying
differentiated leadership roles by boys
and uh rashford and Jack they say the
antagonistic relationship between the
test positive Network and the default
mode Network creates a fundamental
neural constraint on cognition that is
highly relevant relevant to the
different roles and capabilities that
effective leaders must astutely juggle
and deploy an important consequence of
this constraint is that an over emphasis
on task-oriented leadership can prove
deleterious to an organization in
particular when openness to new ideas
people emotions and ethical concerns are
important to success on the other hand
the over emphasis on
relationship-oriented leadership may
prove deleterious to focus and and the
execution of clearly defined goals and I
thought that this was really cool I just
have never like
thought about leadership like this but
it's definitely there's like the people
component and like the GSD component
like getting stuff done
um and and like leadership requires both
and like you can't like you need to have
both and I also thought like that the
ambidexterity here like where Bishop
talks about is like interesting because
like you think about like organizations
and you always hear like the phrase like
the left hand doesn't know what the
right hand is doing and I wonder if like
that's why this is like uh this

different management
um like role or dichotomy is like
labeled like the ambidexterity or like
what what which which came first the
chicken or the egg here
um I don't know to get through any
comments here
one thing I thought was interesting here
is like you know one level there's like
this analogy between like the default
mode Network and the test positive
network with we might find in an
organization
but I've always was interesting too was
thinking about how the individual
relates to the organization and that
like you know the the creativity that an
individual or leader can exercise is
kind of constrained by the antagonistic
relationship between their own test
positive and default mode Network right
and you know it's one thing for people
to say like yes like you need to have
this antagonistic relationship and it
used to be this balance but the people
that's the ability for the organization
to do that is constrained by the people
within that organization their own
individual abilities
to kind of switch between these two
these two systems and you know it's it's
kind of a lot yeah I guess you're
basically like the organization's
ability to do this runs up against the
biology of like the individuals and
their own ability to switch between
these different modes and I think it's
that ability to kind of run these two

modes in parallel is quite rare and it's one thing to ask your leaders to have it but to ask your entire organization to have it it's like a whole much different and much bigger ask and so one of my things I keep thinking about when I'm going through this book is like to what extent does the full organization need to have internalized like Duality management or is it really just the leaders or I think what Bajan would say is like the board but I don't that doesn't seem to connect each level of the hierarchy like it seems like each level of the hierarchy should be able to manage these two different modes of processing but it's a rare skill so how is that even possible to do um yeah I'm not sure if you get any thoughts on that yeah a few more one is the switching which is sometimes even potentially like faster than a second of our own behavior or neurophysiology from task oriented to mind wandering and to what extent we sometimes just zone out thinking daydreaming reflecting integrating looking at our schedule looking at our inbox like it's not task driven but that is like a space that can then transition into a task not always the task we want but it can transition to a task so that's kind of cool and antagonistic's one way to frame it you could also call them synergistic or synergetic because they're part of that

Unity is plural and at minimum to angle where no single neural network could be tuned to be able to be both diffusive and task oriented so why not have some type of oscillatory process that enables those two modes like having a forward in a reverse in a car instead of trying to make one gear that can do two things that are contradictory

yeah it says super good point

um and in this paper they really uh talked about

um they went in a greater detail about the task positive Network and the default mode Network and they said that the task positive network is activated during a broad range of non-social tasks and is thought to be important for problem solving focusing attention making decisions and control of action in other words for getting things done however activation of the task positive Network also has a deleterious effect on other cognitive functions that are essential to leadership it suppresses activity in the default mode Network

and similarly the default node mode network plays a role in emotional self-awareness social cognition and ethical decision making it is also strongly linked to creativity and insightful problem solving and then the activation of the default mode Network they said elsewhere inhibits the the task positive uh Network so I thought that there's like this like neat and it is like you know Unity you need both

um you can't constantly like be focused on a task like you guys know how that feels right like to just be like uh I'm like out of it like you have to have that like you need to like have work and play

um in order for some kind of balance or like it's it is unifying to be balanced

I think

just one thing that's made me remember I think I've actually mentioned the same previous stream but you know in regards to like switching between the two Gears of the car rather than trying to like run them both at the same time one way that this is was phrased to me by one of my previous uh CEOs was you have like art projects and you have science projects and so like your art being your default mode projects your science being your task positive projects and like explicitly saying like we are in art mode right now like don't hold us accountable to any metrics like this is an art project we're exploring uh versus science which is like hold me accountable I mean and that means hold me accountable to metrics like I'm going to set a number and I'm going to hit that number and like that's how you I should be evaluated and so I think there are ways you could probably like frame switching between these modes in an organization

yeah thank you Tyler and one biological angle on this sometimes this type of oscillation is called a winnerless competition and it's modeled using

Predator prey type Dynamics uh Lotka

Volterra is the name of the dynamical formalism

and in the SPM textbook and package from

Carl Friston and colleagues earlier work

there's a lot of examples of winnerless

competition lock of Volterra models

being used to model activity amongst

different brain regions and this

winnerless competition is an interesting

framing on everything from attention to

sleep it's like sleep is not trying to

be a hundred percent of the time

wakefulness is not trying to be 100 of

the time they inhibit each other

until there's a phase transition and

then it's like a dwell in one state and

then

that inhibition shifts and one

oscillates back and so similarly like a

predator doesn't over the long run

really want to eradicate the prey but

their populations fluctuate fluctuate

and that's a winnerless competition and

in fact that's a co-winning competition

and just one last Point

um I remember in the selfish Gene some

of the discussion of

out there on the Savannah look at the

cheetah eating the gazelle how can one

say that that's like a mammal level

selection at play that doesn't make any

sense it's like but actually if that

ecological relationship weren't there

then neither population would have the

resilience

so how we frame things and especially

the level of holism that we bring is

going to set the stage for whether we
see something as sustainable or
unsustainable adversarial or
non-adversarial

awesome

um

okay so moving on uh Bijan says free
energy governance is firmly rooted in
the belief that in a discontinuous and
distributed world the marginal benefit
of structural reorganizations as well as
structural compartmentalization of
exploitation and exploration are rather
sub-optimal or possibly

counterproductive competitiveness and
survival are increasingly determined by
the extent to which a system is rather
less rule or top-down supervised
learning based exploration happens
increasingly in real time within
exploitation in effect the emergence of
the new is essentially hypothesis
testing within the existing and I really
liked this like um uh like Daniel you
had mentioned earlier about like is this
a partition or is this like a nesting
um and like this exploration and
exploitation

um is like is a partition and also is
like a nesting

um based on on what Bijan says here uh
he also says in a distributed and
discontinuous Market environments the
real leadership challenge is to
de-antagonize the Past positive network
default mode Network dichotomy in the
form of Duality management

um do you guys have any comments here

all right

moving on do you want to read these

claims Tyler

sure feg's cross hierarchical

connectivity logically concludes that

managing the tpn DNN antagonism it's not

a top-down challenge calling for

structural compartmentalization but one

of Duality management by vacillating

between or among discrete formal

organization mode such as centralization

and decentralization the organization

may increase dynamically the levels of

exploration and exploitation Beyond

those achievable through an approach

based strictly on a static design Choice

summary

pretty much conceived organizing as an

informally deep learning and

self-designing Compass centralizing

intelligence at any hierarchical level

of the organization can be

counterproductive increasing free energy

promoting entropy and eventually

capitulating to tpn DMS and antagonistic

forces

um so I think one thing you might have

been referencing here is when you're

structurally trying to create you know

your DNN and tpn and like having them

separate from each other this is what's

like traditionally done in companies of

having like an r d department where like

this is where all the like the new ideas

come from

um and so it's I think he was

juxtaposing the fpg approach to like

that very traditional model

oh

um

so we'll move on here so he gives here the free energy governance response to the task positive network default mode Network Challenge and it's this three-fold response and he says sensitizing the entire organization in regard to Upper echelon's tendency to top down extrapolating is dominant supervising knowledge structures logic and past experience as a guide and prediction for future success a fallacy that tasks specific exploitation tends to reinforce embracing strategic renewal as a matter of continuous deep learning in the form of cultivating cross-hierarchical processes empowering purpose-directed and action-generated bottom-up Market stimuli as a basis for double Loop learning question the very models that generate the top-down predictions and three default abductive attentional engagement at every level of the organization forms the basis for Effective top-down bottom-up prediction error minimization after noticing an environmental change it is essential to question one's understanding of how the subsystems of the environment EG technology and markets and the organization EG knowledge and goals are linked without losing sight of the fact that environment is not objectively defined but a product of action generated and socially constructed shared meaning and expectations so I think this is like how the Duality

management works essentially um that's kind of what what I took from here Tyler do you have comments yeah and just that like these are all really big asks to make up an organization at every level like I think it's one thing to say like yeah I agree that this is the answer and if this could be done like be like hell yeah like I'm all for it but I again feel like you're kind of running up against people's individual abilities to manage Duality just within themselves um and that will constrain your ability to do this with an organization foreign because I think about the cognitive cone right like uh when we that we talked about with Michael Levin and um it's funny like what you just said is like the flip of the cognitive cone so usually it's like the you know the the larger structure or like the more top level hierarchical structure will cognitively constrain like things that are nested within it so the shape of that cone the the shape of the individual is within that cone can't um be any other shape right like so they can't come out of the cognitive cone of the entire organization but but it's interesting to think about how um what is happening at the individual level like the shape of the individual cognitive cone also uh lends itself to the configuration of the higher level cognitive cone the energy of comments I

see you're writing furiously no
all right
um drawing figures doodling seriously
um okay so bishan goes on to say talk
about a salience trap he says while
salience is critical to an active
inference it can nonetheless potentially
increase free energy to empower
abductive sense making so that weak
signals can serve as an infrari
information Rich enhancement to
predictions processing listening to it
wants to emerge organizational design is
challenged to neutralize the tendency of
experienced managers and board members
to overconfidently superimpose the
potential fallacy of their historic and
self-confidence-powered pattern
recognition since the human brains task
positive Network and default mode
Network spheres are antagonistically
related the only way to embrace
antagonism as Duality is by linking in
circular causality top-down predictions
and bottom-up stimuli
um and so here it's it I think that um
Bijan is really uh maybe kind of
criticizing or or like the maybe not
criticizing maybe uh
I don't know maybe criticizing's the
right word like how experience leads you
to prioritize like leads you to
um Place importance on things that and
maybe perhaps miss things that are
important in an Adaptive organization
I don't know do you guys have comments
here
we're drawn to the salience not to the

meaningful or adoptive necessarily we
hope that our salience
field can be aligned with what's
adaptive and meaningful but
I think we found out that that is not
always the case
and it's funny like I think about just
in scientific discovery right like how
we're so attuned we're looking for the
signal that we're looking for
um it takes like a very perceptive and
Adept scientist to not fall into that
salience trap and perhaps notice
um what's going on in the periphery
right like even though this is the
center this is what we're focusing on
there's a lot of peripheral things
happening that that might actually be
important or as important or maybe more
important than than the central thing
that we're focused on
um yeah I mean I guess that kind of the
other thing I say is that this is just
like another
it's one of those things where I agree
with this in Prince and like
theoretically but
there's so much stress going to like
many of these decisions like for example
if you have to do layouts for example
like it's very time sensitive and the
stakes are really high and so you know
going off your previous experience can
be really helpful in there because you
have to just make it really quick
decision and it's often like going to be
heuristic space uh which may be not
necessarily optimal but you really are

often constrained by like your time and resources and like really processing all of your bottom up seeing the signals like even with the best intentions is really hard to do

um and so again it's not to say like you shouldn't try to do it or it can't be done but it's just like

it's another one of these things we're like hey this is a really big ass and I just want to call out like this is really really really really really hard to do especially when you're under under the gun

yeah let's see this is in the ideal world this is how it would work right

um okay do you want to read this claim first Tyler

sure

um all right so indeed this real-time switching vacillation process cannot be relegated to a single person if you need a CEO or executive chairman or upper echelon as a team but must be the result of an unsupervised and FTP determined learning process that relates to top-down predictions and bottom of data input circular causality to vacate resource reconfiguration accordingly consequently the board's potentially unique dmn Dimension cannot be ignored but must be integrated into the process at the very top the board of directors has given us cognitive diversity and composition and cognitive distance firms done at tpn exploitation Logics emerges as a monitoring surveyor of the system that is centered around managing the

dynamic interplay of dual forces
exploitation exploration and financial
and strategic controls
um yeah yeah something I thought that
was interesting here is that like
you know while he's making this claim
and which he's been through the whole
book that like pay the boards in this
really neat position to be just kind of
like an integrated unit
it kind of comes back to what we were
saying earlier of like okay who was
actually doing this do you want to be
management and so in one sense I'm like
okay it makes sense that the board would
be doing this because you can hold them
accountable to holding these different
ideas in their head at the same time uh
managing Duality but it seemed I would
think that you would need that Duality
management to be going down the
hierarchy not just at the border as well
and at each level the hierarchy is
managing duality in some sense
um
not from so I very much want to ask you
on that when we uh got on a call with
him
blue day one thing to add here
no
um but this is one sentence that I
pulled out randomly
um from the chapter that just like
struck me and like he just like threw it
in here and didn't unpack it through
anything the firm is the generative
social field and emerges in language and
I thought that this this was like really

profound so

um I don't know I just put it in here

because like just like wow that's very
deep

um I don't know

I don't know if you guys have comments

or this is something like I would ask

John to expand upon

you could ask a hundred people with
different backgrounds

what this invokes for them

and see where that goes

that sounds like a fun project it was

just like a like a sentence that seemed

very like yeah deep and like just

casually thrown in there yeah so I'm

gonna get this tattooed on my chest

right put it on your next business card

that'd be awesome that'd be an awesome

like statement for a business card right

um that'd be great

so what is it you say you do

I don't know I guess that's true this is

the business philosophy

awesome

so do you guys have any final like

comments questions Tyler did you tell us

all the things you wanted to tell us

no that's it for me

okay

yeah we're moving along through the

structure I'm looking forward to more

conversations and also some digestions

and Reflections making some schemes to

visualize at a high level what we've

been coming away from because this is

like our eighth session and I think

we're starting to see his patterns and

style of laying out what he wants to
convey and integrate

yeah I definitely think we're getting
like the repeat like repeat the message
and so so it's starting to drive some of
it home for me at least so coming up to
the end of the second part of the book
next week and I'll be looking forward to
that and um to hopefully discuss soon
with the author at his convenience that
that part

all right well thank you blue and Tyler
farewell

bye bye

next week same time